

Applicant Profile & Supplementary Information

Ismail Aissaoui
<https://ismail-aissaoui.vercel.app>

Links to Code and Theses

- **GitHub Repository:** <https://github.com/IsmailAissaoui>
- **Master's Thesis (Cardiovascular Medical Digital Twin):** Available via Academic Portfolio
- **State Engineer Thesis (Gas Turbine Digital Twin):** Available via Academic Portfolio

Record of Academic Courses & Grading

Institution: Ecole Supérieure en Informatique de Sidi Bel-Abbès (ESI-SBA)

System: 0-20 scale (10 is passing; >14 is Very Good; >16 is Excellent)

- **Core Mathematics & Theory:** Calculus, Linear Algebra, Probability & Statistics, Optimization Theory.
- **Computer Science & AI:** Machine Learning, Deep Learning, Computer Vision, Signal Processing, Data Structures & Algorithms.
- *Note: Full official academic transcripts and translated degree certificates are available upon request or upon formal enrollment.*

Key Academic Projects

1. Cardiovascular Medical Digital Twin (Master's Thesis)

Contribution: Engineered "Medical_MambaTS", a linear-time state-space model for sub-millisecond bedside inference. Developed PI-DeepONet, a Physics-Informed Operator Network that explicitly maps 0D Windkessel differential equations into the neural architecture via PyTorch Autograd. Designed a "Missingness-Aware Gate" to dynamically handle sensor dropouts in ICU data.

2. Gas Turbine Digital Twin (State Engineer Thesis)

Contribution: Built predictive maintenance architectures using ST-KDFormer and Temporal Convolutional Networks to predict thermodynamic degradation and sensor drift in complex rotating machinery. Validated against real-world, high-frequency SCADA telemetry.